AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION 2010

General Mathematics Paper I

Time allowed: 40 minutes Marks 30

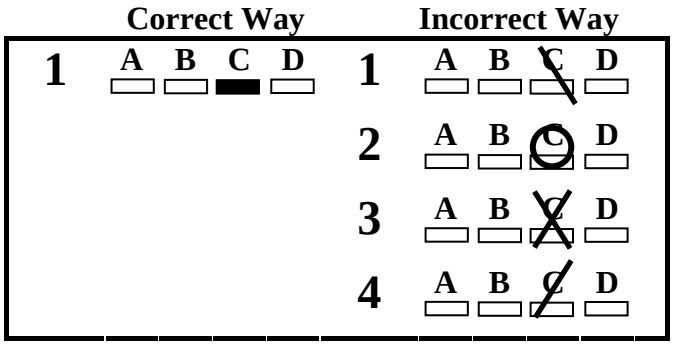
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

7. You may use a simple calculator if you wish.

1. If $a + b = 2$ and $a - 2 = 0$, then $a^2 + b^2$ is equivalent to

   A. 8
   B. 6
   C. 4
   D. 2

2. Which one of the followings is a binomial surd?

   A. $\sqrt{5}$
   B. $\dfrac{\sqrt{7}}{\sqrt{3}}$
   C. $\sqrt[3]{5} - 1$
   D. $\sqrt[3]{4}$

3. An algebraic expression for which $P(1) = P(-1)$ is

   A. $P(x) = x^2 - 1$
   B. $P(x) = x - 1$
   C. $P(x) = \dfrac{1}{x-1}$
   D. $P(x) = \dfrac{1}{x^2 - x}$

4. The factorized form of $2a^3 - 4a^2 + a$ is

   A. $a(2a^2 - 4a)$
   B. $2a(a^2 - 2a)$
   C. $a(a^2 - 2a + 1)$
   D. $a(2a^2 - 4a + 1)$

5. The remainder of $\left(1 - 2x + x^2\right) \div (1 - x)$ is

   A. –1
   B. 0
   C. 2
   D. 4

6. If $(x + 1)$ is a factor of $P(x)$, then the second factor of $P(x)$ is

   A. $\dfrac{x+1}{P(x)}$
   B. $\dfrac{P(x)}{x+1}$
   C. $(x+1) \times P(x)$
   D. $(x+1) + P(x)$

7. The highest common factor (HCF) of $a + b$ and $a^2 + 2ab + b^2$ is

A. $a^2 + b^2$
B. $a^3 + b^3$
C. $(a+b)^3$
D. $(a+b)$

8. The square root of $x^2 + \frac{1}{x^2} - 2$ is equal to

A. $\pm\left(x + \frac{1}{x}\right)$
B. $\pm\left(x - \frac{1}{x}\right)$
C. $\pm\left(x + \frac{1}{x} - \sqrt{2}\right)$
D. $\pm\left(x + \frac{1}{x} - 1\right)$

9. $\frac{1}{x} - \frac{1}{x(x-1)}$ is equal to

A. $\frac{1}{x-1}$
B. $\frac{1}{x(x-1)}$
C. $\frac{x}{x(x-1)}$
D. $\frac{x-2}{x(x-1)}$

10. Akber bought 1 litre milk and 1 dozen eggs. The price of 1 litre milk is Rs. 50. If he paid Rs.122, then the price of one egg was

A. Rs. 6.0
B. Rs. 6.8
C. Rs. 7.0
D. Rs.5.5

11. The solution set of $1 + \sqrt{1-x} = 0$ is

A. {0}
B. { }
C. {1}
D. {0, 1}

12. If $a + x = bx$, then x is equal to

A. $\dfrac{b}{a-1}$

B. $\dfrac{a}{b-1}$

C. $\dfrac{a}{b+1}$

D. $\dfrac{b}{a+1}$

13. If the solution set of the equation $x^2 - x + p = 0$ is $\{-2,3\}$, then the value of p is

A. -6
B. -1
C. 1
D. 6

14. One of the roots of $x^2 - 2x + 1 = 0$ is

A. -1
B. 1
C. 2
D. -2

15. If $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, then which one of the followings is NOT possible?

A. $A \times B$
B. $A \times A$
C. $B \times A$
D. $C \times C$

16. If $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \times R = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, then the order of matrix R is

A. 2×2
B. 2×1
C. 1×2
D. 1×1

17. The multiplicative inverse of $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ is

A. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

B. $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

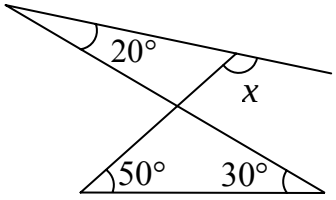
C. $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

D. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

18. The inverse of the matrix $\begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix}$ does NOT exist if

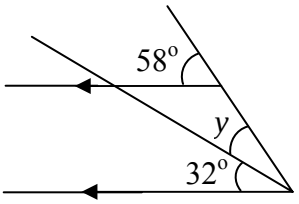
A. $x > 1$
B. $x = 0$
C. $x = -1$
D. $0 < x < 1$

19. In the figure, the value of x is



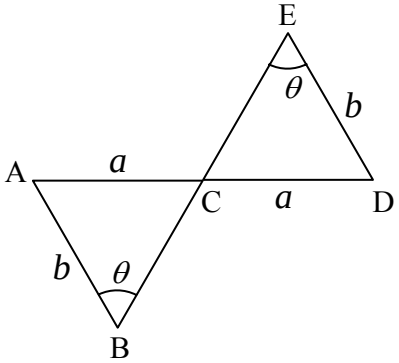
A. $100°$
B. $110°$
C. $120°$
D. $130°$

20. In the figure, the value of y is



A. $13°$
B. $26°$
C. $32°$
D. $29°$

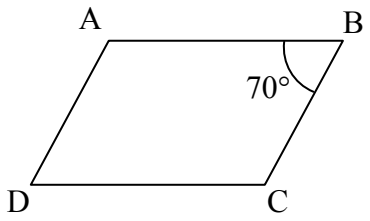
21. In the figure, $\Delta ABC \cong \Delta DEC$ because their corresponding



A. angles are congruent.
B. two sides and one angle are congruent.
C. two angles and one side are congruent.
D. two sides and included angle are congruent.

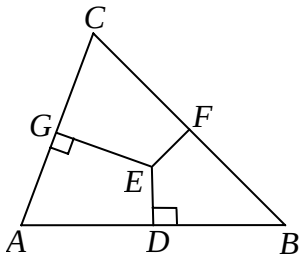
22. In the parallelogram ABCD if $m\angle B = 70°$, then $m\angle A + m\angle C$ is equal to

A. $140°$
B. $160°$
C. $180°$
D. $220°$



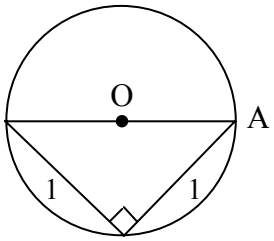
23. In the triangle, if *D*, *G* and *F* are the mid points of *AB*, *AC* and *BC* respectively, then *EF* is the

A. median of BC.
B. altitude of BC.
C. right bisector of BC.
D. angle bisector of angle A.



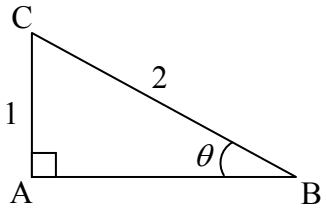
24. In the figure, radius OA of the circle is

A. 1
B. $\frac{1}{\sqrt{2}}$
C. $\sqrt{2}$
D. 2



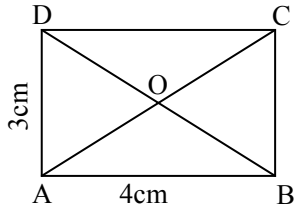
25. In the right angled triangle, the length of AB is

A. 1
B. $\sqrt{2}$
C. $\sqrt{3}$
D. 2



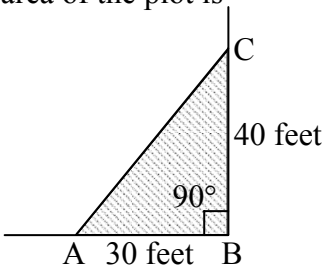
26. In the rectangle *ABCD*, the $m\overline{OB}$ is equal to

A. 7cm
B. 5cm
C. 4cm
D. 2.5cm



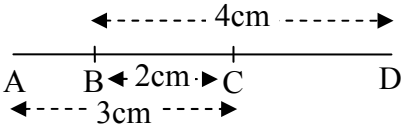
27. The figure represents a triangular plot ABC. The area of the plot is

A. $120\ ft^2$
B. $600\ ft^2$
C. $1200\ ft^2$
D. $2500\ ft^2$



28. In the figure A, B, C and D are collinear points. If $m\overline{AC} = 3cm$, $m\overline{BD} = 4cm$ and $m\overline{BC} = 2cm$, then the length of $m\overline{AD}$ is

A. 5 cm
B. 6 cm
C. 7 cm
D. 8 cm



29. A right angled triangle may be

I an equilateral triangle
II an isosceles triangle
III a scalene triangle

A. III only
B. I and II only
C. II and III only
D. I, II and III

30. The shortest distance between (2, 0) and (0,–2) is

A. 2
B. $2\sqrt{2}$
C. 4
D. 8

END OF PAPER

Please use this page for rough work